SIP One Minute Speech

Caleb Quisenberry | SIP408 | September 27, 2026

We rely on tower-mounted radios for internet access at home, work, and school. When a tower or radio loses power, customers can lose that connection. What if technicians could identify the problem before climbing the tower?

I'm Caleb Quisenberry, studying Network Engineering and Cybersecurity. My project is the Intelligent Tower Mounted Distribution Node.

My innovation combines power and fiber distribution with planned wireless fault reporting to simplify installation and troubleshooting.

I completed two field installations. After changing jobs, I lost access to them and planned a 12-volt Arduino and LoRaWAN prototype. This will allow me to explore a network format that I have not worked with, and the associated security protocols.

For the demonstration, I'll use a switch to turn off its main supply. UPS based backup power will keep the controller and sensors running to report the outage.

My Week 4 target is assembly, testing, and recording, depending on parts availability.

Easier installation and faster troubleshooting can help providers keep your internet connected. Thank you.

Opening visual: Show tower photographs, including an existing elevated view and a technician on a tower.

Demonstration visual: Once built and tested, show normal operation, unplug the main 12V supply, show the received outage report, then reconnect power and show recovery. Keep the gateway powered separately.

Until the prototype works, show the design and label the demonstration as planned.